

Pre-fit Audit

Diagnostics computed from raw data only, before any model fitting

Every observation-model and temporal-structure choice is fixed here, from the data alone, so that those choices are pre-registered rather than selected after seeing a fit. Decision rules are stated with each test.

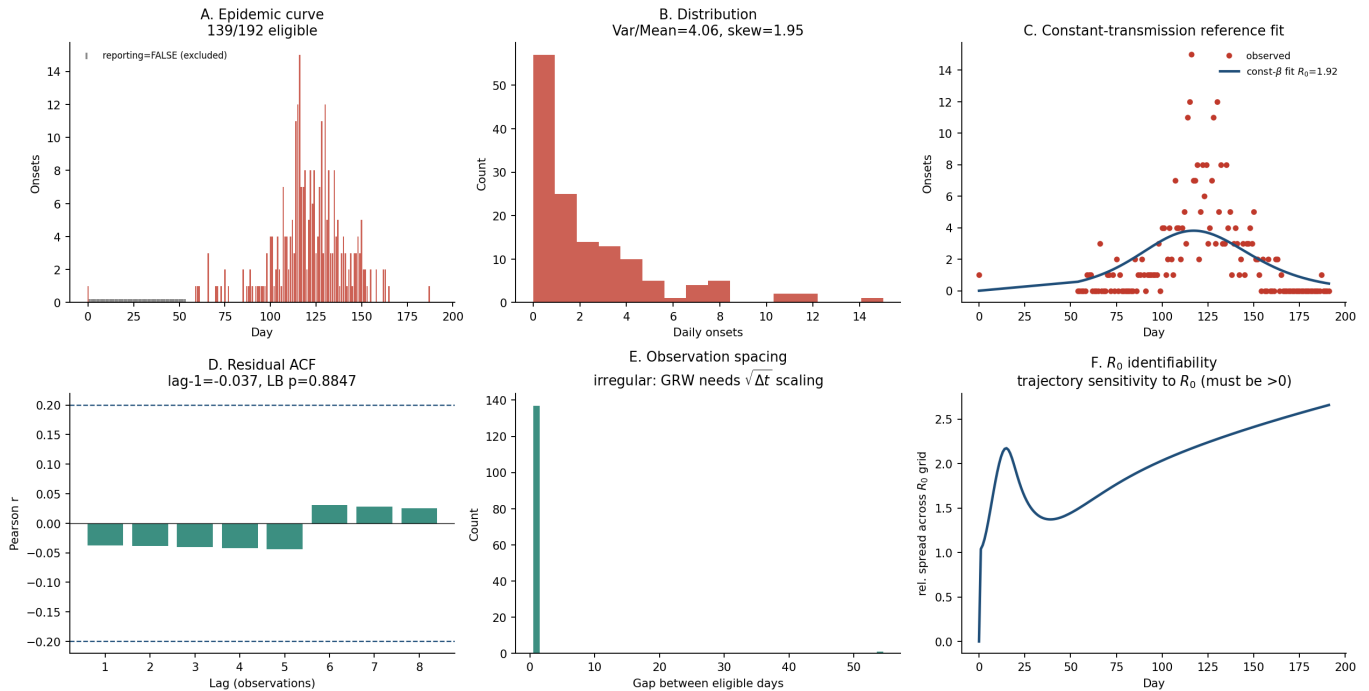
1. Summary of decision rules

Disease	Var/Mean	Skew	Zero frac	Residual lag-1	Ljung-Box p	GRW warranted?
ebola	4.06	1.95	0.41	-0.037	0.885	no
measles	10.64	3.31	0.59	-0.005	0.731	no
influenza	103.64	0.45	0.00	+0.796	0.003	YES

Rule: a GRW on $\log \beta(t)$ is warranted only if the residuals of the constant-transmission mechanistic fit show $|\text{lag-1 ACF}| > 0.20$ or Ljung-Box $p < 0.05$. Ljung & Box (1978) *Biometrika* 65(2):297-303.

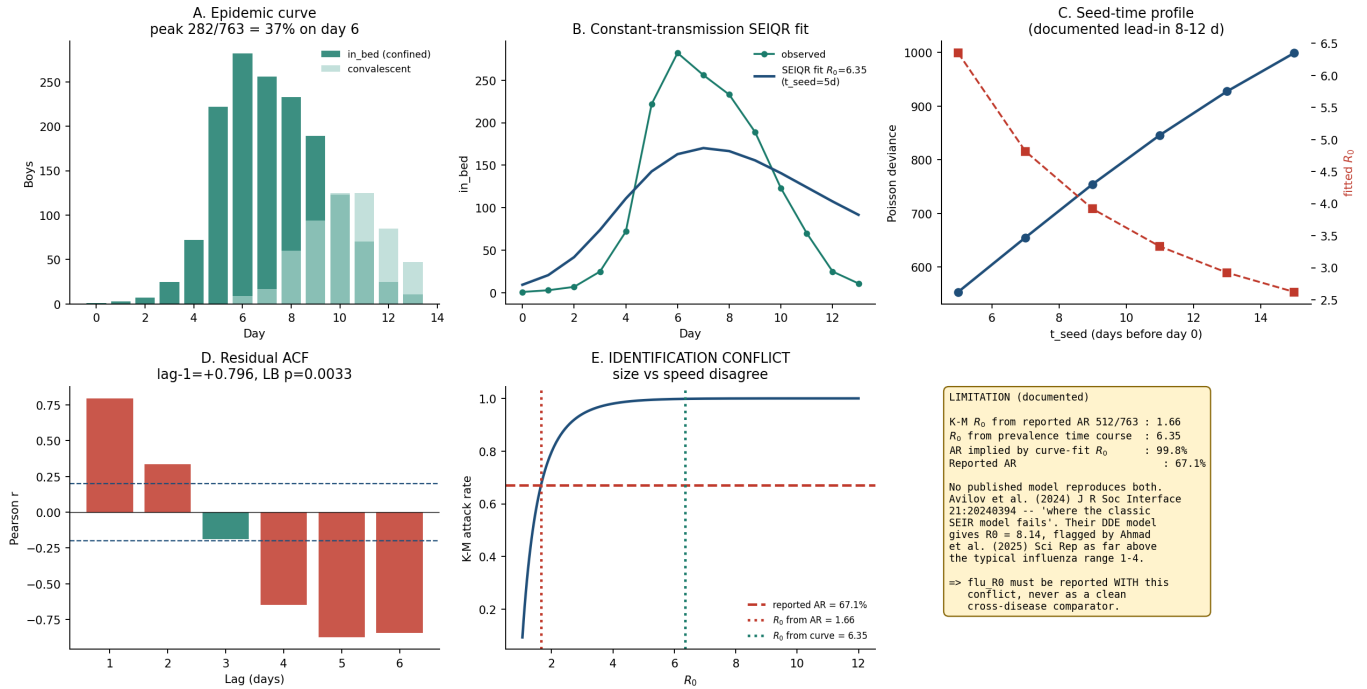
Ebola Kikwit 1995

Ebola Kikwit 1995 - Pre-fit Audit



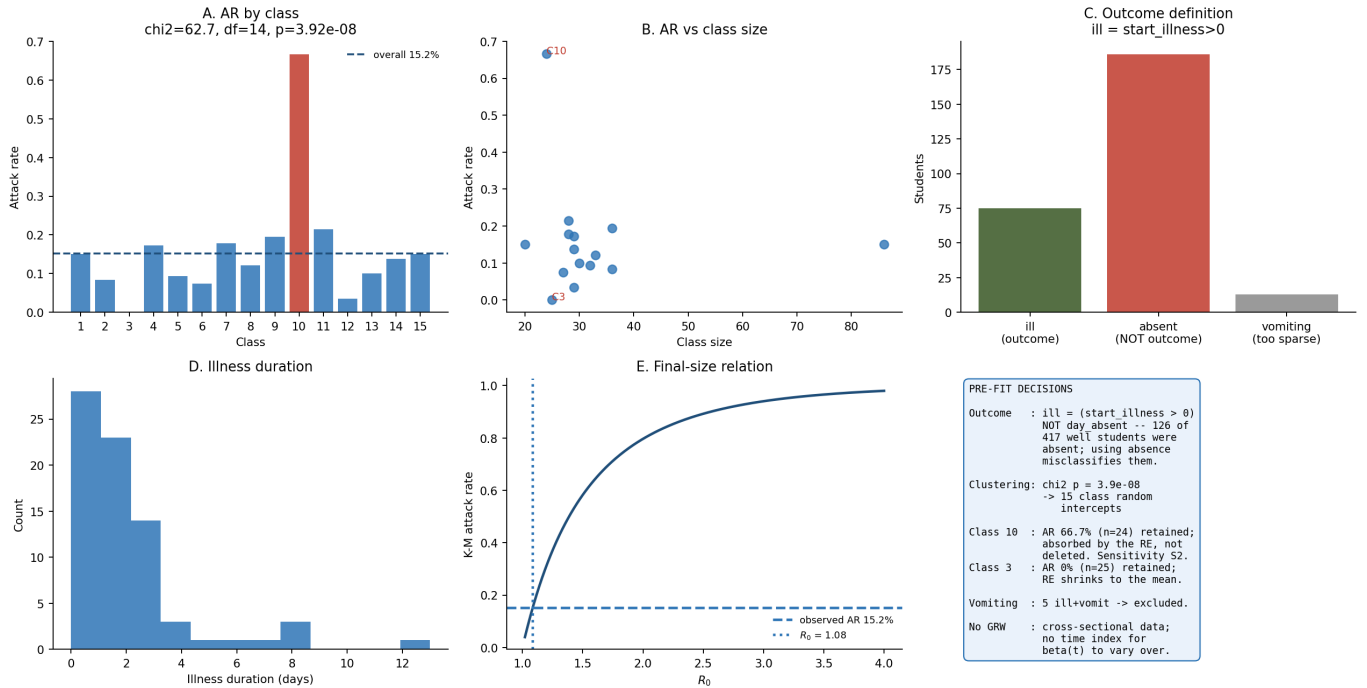
Influenza England 1978

Influenza England 1978 - Pre-fit Audit (SEIQR, in_bed = confinement compartment)



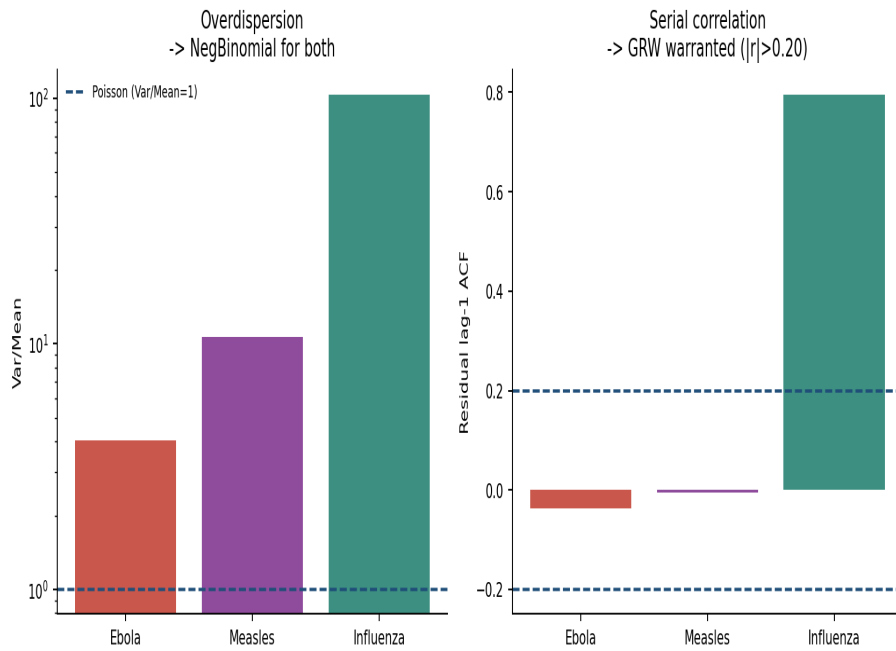
Norovirus Derbyshire 2001

Norovirus Derbyshire 2001 - Pre-fit Audit



Cross-disease overview

Cross-disease pre-fit overview



OBSERVATION MODEL BY ARM	
Ebola	onset = incidence flux sigma*E(t); NegBinomial 139/192 eligible days
Measles	rash onset = incidence sigma*E(t); NegBinomial 86 days, census cohort
Influenza	in bed = confinement Q(t) SEIQR; NegBinomial NOT I(t) -- see Avilov et al. (2024)
Norovirus	ill = binary per student Bernoulli + 15 class REs R0 via K-M final size
Zombie	prior predictive only; no data, no likelihood

Influenza identification conflict

R_0 implied by the reported attack rate $512/763 = 67.1\%$ is 1.66. R_0 implied by fitting the prevalence time course is 6.35, which in turn implies an attack rate of 99.8%. These cannot both hold. Avilov KK et al. (2024) J R Soc Interface 21(220):20240394 show that no classic compartmental model reproduces both the time course and the final size for this dataset. See docs/INFLUENZA_1978_LIMITATION.md.